Micro Set 3. Negative Externalities — Singapore Carbon Tax and Vehicle Emissions

Instructions. This booklet contains a single case study on negative production and consumption externalities, followed by six short-answer questions ranging from 2 to 6 marks.

Section Case Study. Singapore's Carbon Pricing and Vehicle Emissions Policy

Table 1.1: Singapore Carbon and Vehicle-Related Indicators, 2019–2025

Indicator	2019	2020	2021	2022	2023	2024	2025
Carbon Tax (S$ per tCO_2e)	5	5	5	5	5	25	25
Industrial Carbon Emissions Index (2019=100)	100.0	96.2	99.4	97.8	96.1	92.5	—
Vehicle Emissions Scheme (VES) Surcharge max (S$)	20000	25000	25000	25000	25000	25000	25000
VES Rebate max for cleanest band (S$)	20000	25000	25000	25000	25000	25000	25000
Petrol-fuelled Car Population (000s)	405	410	408	401	384	367	—
Electric Vehicle Share, new registrations (%)	0.2	0.9	4.6	11.7	18.1	33.6	—
PSI Particulate Matter ($PM_{2.5}$, annual avg, $\mu g/m^3$)	15	13	11	14	13	12	—

Source: Adapted from National Environment Agency, Land Transport Authority, and Ministry of Sustainability and the Environment.

Extract 1.1: Internalising the externality: carbon pricing and vehicle emissions in Singapore

Singapore's framework for addressing negative externalities in production and consumption has been incrementally tightened across the past decade. The carbon tax, introduced in 2019 at S$5 per tonne of carbon dioxide equivalent (CO_2e), covers approximately 80 per cent of national greenhouse gas emissions, applying to direct emitters above the 25 $ktCO_2e$ per year threshold — principally power generation, refining, and large chemical and semiconductor manufacturers. The tax was raised to S$25 in January 2024 and is scheduled to rise to S$45 by 2026 and S$50–80 by 2030.

"The carbon tax is the most economically efficient way to drive emissions reductions because it makes polluters pay the social cost of their emissions, incentivising them to either reduce emissions or pay," a Ministry of Sustainability and the Environment spokesperson said. "Critically, it preserves firms' freedom to choose how to abate, allowing each firm to identify the lowest-cost path to lower emissions."

Industrial emissions have fallen modestly since 2019, although the response so far has been muted: a 5 per cent decline in five years against a target of 36 per cent by 2030. Several factors contributed: the initial S$5 rate was significantly below the marginal abatement cost for most heavy emitters, abatement options for the chemicals and refining sectors require substantial capital investment with long lead times, and firms have rationally awaited clarity on the long-term trajectory before committing major investment. The 2024 increase to S$25 marks the first material price signal.

The Vehicle Emissions Scheme (VES) for private cars works through a different mechanism: a feebate that imposes a surcharge of up to S$25000 on the most polluting vehicles and offers a rebate of up to S$25000 for the cleanest band, primarily electric vehicles (EVs). Combined with the Early Adoption Incentive (EAI) for EVs and the planned phase-out of internal combustion engine vehicle registrations by 2030, the policy has driven EV share of new registrations from 0.2 per cent in 2019 to 33.6 per cent in 2024. “The VES has been substantially more effective at changing consumer behaviour than the carbon tax has been at changing producer behaviour,” an environmental economist at NUS told the Business Times. “This reflects both the directness of the price signal and the fact that consumers face simpler, more visible trade-offs than firms making multi-year capital decisions.”

Singapore’s overall $PM_{2.5}$ readings remain influenced not just by domestic emissions but by transboundary haze from regional forest fires and Southeast Asian industrial activity — an externality that cannot be addressed through domestic policy alone. The annual average has hovered between 11 and 15 $\mu g/m^3$, above the World Health Organization’s 5 $\mu g/m^3$ guideline but within Singapore’s own air-quality targets.

Source: Adapted from Ministry of Sustainability and the Environment, National Climate Change Strategy 2050, Land Transport Authority Annual Report 2023, and The Business Times, 4 March 2024.

Question 1 **[2 marks]**

Define “negative externality in production” and identify one piece of evidence from Extract 1.1 that industrial carbon emissions in Singapore constitute one.

Question 2 **[2 marks]**

With reference to Table 1.1, describe the trend in electric vehicle share of new registrations between 2019 and 2024.

Question 3 **[4 marks]**

Using evidence from Extract 1.1, explain why the carbon tax at S$5 per tCO_2e (2019–2023) produced only a 5 per cent reduction in industrial emissions over five years.

Question 4 **[4 marks]**

Explain how the Vehicle Emissions Scheme rebate of up to S$25000 for cleanest-band vehicles works as a mechanism to internalise the negative externality from vehicle emissions.

Question 5 **[6 marks]**

With the aid of a fully labelled diagram, explain how a Pigouvian tax set equal to the marginal external cost would correct the negative production externality from industrial carbon emissions.

Question 6 **[6 marks]**

Using evidence from Extract 1.1, explain why the Vehicle Emissions Scheme has been more effective than the carbon tax at delivering measurable behavioural change.

End of Set 3 — Questions